Ocean FCL Multi-Commodity Routing

MVP Formal Model — Phase 1, v2

AI Freight Agent

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Abstract

Formal specification of the optimization model for the ocean FCL multi-commodity routing problem. The model is a Binary Multi-Commodity Flow problem on a commodity-specific subgraph derived from the physical freight network (see `docs/ocean_fcl_multi_shipment_graph.drawio`). Scope: FCL only, FEU+TEU containers, string-based carrier allocation capacity, deterministic time windows. Probabilistic objectives, hazmat/OOG cargo differentiation, and time-phased capacity management are explicitly deferred to P1.

Contents

1 Problem Statement

Given a set of shipment requests (commodities), each defined by an origin location, destination location, cargo volume, cargo weight, and time constraints, find a minimum-cost assignment of commodities to paths through the multimodal freight network such that:

- every commodity is delivered on or before its deadline,
- no ocean sailing exceeds its vessel slot capacity,
- no carrier service string exceeds its contracted forwarder allocation in any period, and
- optional per-commodity budget caps are respected.

MVP scope.

- FCL only; LCL deferred (PRD Open Question 8).
- Container types: FEU (40′HC, $\approx$ 76 CBM usable, $\approx$ 26,500 kg payload) and TEU (20′, $\approx$ 33 CBM usable, $\approx$ 24,000 kg payload). Optimizer selects mix.
- Port dwell: fixed port-specific mean. Commodity-specific inspection model deferred to P1.
- Carrier allocation: string-level monthly contracted block, with current utilization tracked. Time-phased capacity release deferred to P1.
- Objective: minimize total freight cost subject to deterministic time-window feasibility. Probabilistic objectives deferred to P1.
- No transshipment in MVP. Single-leg ocean sailing only.

2 Graph Structure

2.1 Physical Network Graph $G(N, A)$

The freight network is a directed graph $G = (N, A)$.

$$N = N_O \cup N_{\text{POL}} \cup N_{\text{POD},A} \cup N_{\text{POD},E} \cup N_D$$

N_O — **Origins** One per shipper origin door. Commodity source nodes.

$N_{\textbf{POL}}$ — **Ports of Loading**
Container terminals (e.g., SHA, NGB, SZX). Shipments arrive by pre-carriage, depart on vessel sailings.

$N_{\textbf{POD},A}$ — **POD Arrival**
One per discharge port (e.g., LAX_A, LGB_A, SEA_A). Vessel arrives; container cannot depart until dwell arc traversed.

$N_{\textbf{POD},E}$ — **POD Exit**
One per discharge port (LAX_E, LGB_E, SEA_E). Customs cleared; container available for inland pickup.

N_D — **Destinations**
One per delivery door. Commodity sink nodes.

$$A = A_{\text{PC}} \cup A_{\text{OC}} \cup A_{\text{DW}} \cup A_{\text{IL}}$$

$A_{\mathbf{PC}}$ — **Pre-carriage**
: (i, j), $i \in N_O$, $j \in N_{\text{POL}}$. Truck from origin door to container terminal.

$A_{\mathbf{OC}}$ — **Ocean sailings**
: (i, j), $i \in N_{\text{POL}}$, $j \in N_{\text{POD},A}$. One arc per scheduled carrier sailing (specific vessel, departure date, trade lane).

$A_{\mathbf{DW}}$ — **Dwell**
: (p_A, p_E), one per POD. Vessel unloading plus customs clearance.

$A_{\mathbf{IL}}$ — **Inland**
: (p_E, h), $p_E \in N_{\text{POD},E}$, $h \in N_D$. Truck or intermodal rail, POD exit to destination door.

Remark 1 (POD node expansion)**.** Each physical POD is split into two nodes — arrival (p_A) and exit (p_E) — connected by a dwell arc. This makes port processing time an explicit model element and provides the structural hook for the P1 commodity-specific customs inspection model (Section 13).

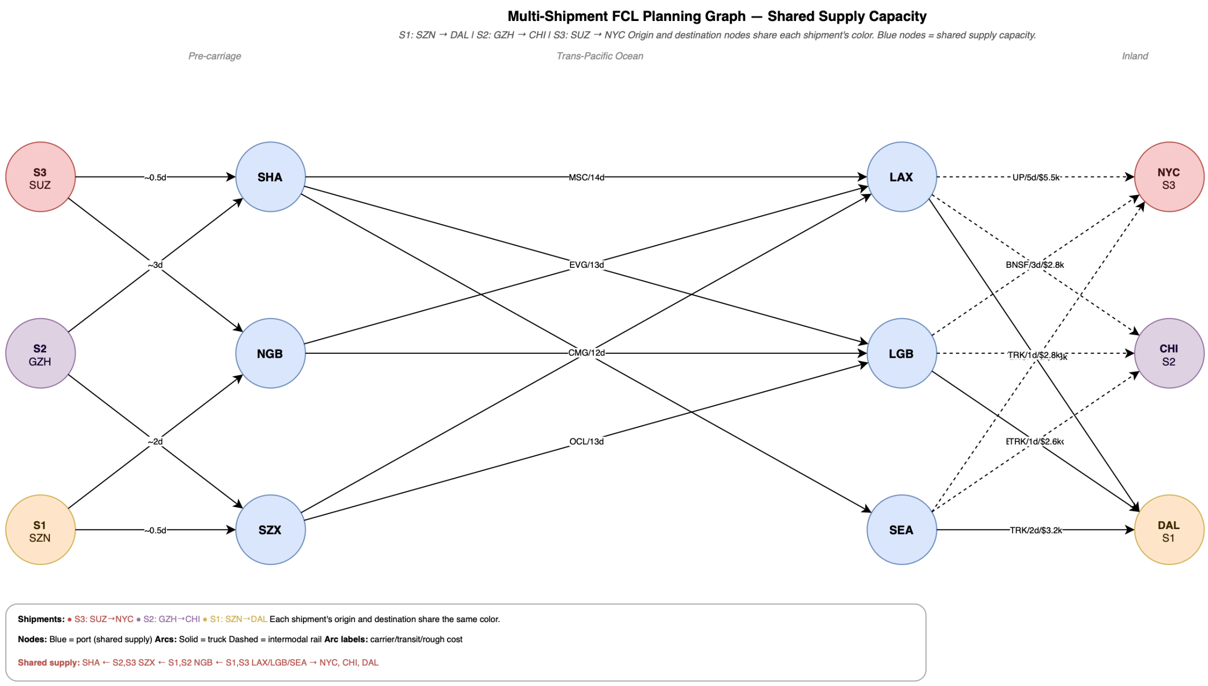

Figure 1: Multi-shipment freight network $G(N, A)$.

2.2 Commodity-Specific Subgraph $G(N_k, A_k)$

The MILP for commodity k operates on a subgraph $G(N_k, A_k) \subseteq G(N, A)$ containing only arcs that (i) are time-feasible for commodity k and (ii) lie on at least one complete path from $o(k)$ to $d(k)$. See Section 5 for the construction algorithm.

3 Sets and Indices

Symbol	Description
K	Set of commodities, indexed by k
N, A	All nodes and arcs
A_{PC}, A_{OC}, A_{DW}, A_{IL}	Arc subsets by type
$A^k \subseteq A$	Feasible arc set for commodity k (Section 5)
$A^k_S \triangleq A^k \cap A_S$	Feasible arcs of type S for commodity k
S	Set of carrier service strings (e.g., MSC Tiger, CMA CGM AEX-1)
T	Set of allocation time periods (monthly)
$P(s)$	Feasible (POL, POD) port pairs covered by string s
N_k	Nodes incident to A^k: $N_k \triangleq \{n : \exists\,(n,j) \in A^k \text{ or } \exists\,(i,n) \in A^k\}$. Used to eliminate vacuous flow constraints in P.1.
$\tau_k(i)$	Earliest arrival time of commodity k at POL i: $\tau_k(i) \triangleq t_k^{\text{rdy}} + \mu^{\text{PC}}_{o(k),i}$. Defined here for reference; computed in subgraph pre-filter step 1 (Section 5).

4 Parameters

4.1 Commodity Parameters

Parameter	Symbol	Description	Unit
Volume	v_k	Cargo volume	CBM
Weight	w_k	Gross weight	kg
FEU-equivalent count	n_k	See Eq. (1); pre-computed	—
Cargo ready	t_k^{rdy}	Earliest pickup from origin door	datetime
Deadline	T_k^{dead}	Latest acceptable arrival at destination	datetime
Origin	$o(k)$	Source node $\in N_O$	—
Destination	$d(k)$	Sink node $\in N_D$	—
Cargo type	τ_k	General / hazmat / temp-controlled / OOG	categorical
HS code	hs_k	6-digit Harmonized System code (P1 customs model)	—
Incoterm	inco_k	EXW / FOB / CIF / DDP	—
Budget cap	B_k	Hard cost ceiling; ∞ if not set	USD

Container count.

$$n_k \triangleq \max\left(\left\lceil \frac{v_k}{76} \right\rceil,\ \left\lceil \frac{w_k}{26{,}500} \right\rceil\right) \tag{1}$$

This accounts for both volume (76 CBM per 40′HC FEU) and weight (26,500 kg payload limit per FEU). n_k is the minimum number of FEU-sized moves for land transport; the optimizer may substitute some FEUs with TEUs on ocean arcs to reduce cost (Section 4.7).

Container specifications.

Container type	Code	TEU slots	Usable volume	Payload	Ocean freight rate
20′ standard	TEU	1	33 CBM	24,000 kg	$\rho \cdot c_{ij}^{\text{FEU}}$, ρ lane-specific (see §4.3)
40′ standard	FEU (std)	2	67 CBM	26,500 kg	c_{ij}^{FEU} − \$50–\$200 vs. HC
40′ High Cube	FEU (HC)	2	76 CBM	26,500 kg	c_{ij}^{FEU} (baseline; defined in §4.3)

MVP uses the 40′HC as the FEU type (76 CBM, 26,500 kg). The 40′ standard is deferred to P1. TEU-only routing for heavy, low-volume cargo is also P1.

Container rate ratios. The TEU/FEU cost ratio $\rho = c_{ij}^{\text{TEU}}/c_{ij}^{\text{FEU}}$ varies by trade lane. Historical averages 2020–May 2024 (Xeneta[1]):

Trade lane	Route	ρ (avg)	ρ (range)
TPEB	China – US West Coast	0.79	0.71–0.86
FEWB	Shanghai – Antwerp	0.56	0.52–0.61
TAWB	N. Europe – US East Coast	0.78	0.68–0.94

Spot ratios fluctuate with equipment balance and demand cycles; on Far East – North Europe, ρ fell to 0.58 at peak demand (July 2024).[2] FEU (HC) vs. FEU (std) ocean freight premium: \$50–\$200 per container, reflecting the 13% volume advantage of the HC over the standard 40′.[3] FEU (std) is not separately modelled in MVP.

Remark 2 (Cargo type in MVP)**.** τ_k is recorded for all commodities but is not used in MVP constraints. Hazmat, OOG, and reefer cargo are routed identically to general cargo in MVP. The instance generator should restrict generated commodities to general cargo unless explicitly testing cargo-type extensions.

4.2 Pre-Carriage Arc Parameters

For each $(i, j) \in A_{\text{PC}}$:

Parameter	Symbol	Description	Unit
Cost	c_{ij}^{PC}	Truck rate per container move. Scales by n_k (one truck per container).	USD/move
Mean transit	μ_{ij}^{PC}	Mean days, origin door to terminal gate	days
Std dev	σ_{ij}^{PC}	Std dev of transit time	days
Road distance	$\text{dist}_{ij}^{\text{PC}}$	Road distance (informational; used by generator)	km

4.3 Ocean Sailing Arc Parameters

For each $(i, j) \in A_{\text{OC}}$:

[1] https://www.xeneta.com/blog/beware-a-potential-flaw-in-index-linked-ocean-container-shipping-contracts-whic
[2] https://www.xeneta.com/blog/dramatic-increases-in-feu-and-teu-ocean-container-market-spreads-key-considerat
[3] https://www.icontainers.com/help/40-foot-high-cube-container/

Parameter	Symbol	Description	Unit
ETD	ETD_{ij}	Estimated time of departure from POL i	datetime
CY cutoff	CYC_{ij}	Latest container arrival at terminal. $\mathrm{CYC}_{ij} \leq \mathrm{ETD}_{ij}$; typically $\mathrm{ETD} - 4\,\mathrm{d}$	datetime
Booking cutoff	BKG_{ij}	Latest booking confirmation with carrier. *[Note C2: Not enforced in MVP. P1: add as a hard pre-filter — exclude any sailing where BKG_{ij} has passed at solve time.]*	datetime
FEU rate	r_{ij}^{FEU}	Base ocean freight rate per 40′HC container	USD/FEU
TEU rate	r_{ij}^{TEU}	Base rate per 20′ container. Typically 0.56–0.86 $\times\, r_{ij}^{\mathrm{FEU}}$ (see rate ratio table)	USD/TEU
BAF	BAF_{ij}	Bunker adjustment per container (same fraction applied to FEU and TEU rates)	USD
PSS	PSS_{ij}	Peak Season Surcharge per container; zero if not applicable. *[Note H5: Not in MVP cost formula. P1: add $+PSS_{ij}$ to the c_{ij}^{FEU} derivation below and to the instance generator.]*	USD
THC (POL)	$\mathrm{THC}_{ij}^{\mathrm{POL}}$	Terminal handling at origin per container	USD
THC (POD)	$\mathrm{THC}_{ij}^{\mathrm{POD}}$	Terminal handling at destination per container	USD
Vessel capacity	$\mathrm{cap}_{ij}^{\mathrm{TEU}}$	Per-sailing slot cap (constraint P.2). MVP proxy: $\alpha \cdot \mathrm{alloc}(s_{ij}, t_{ij})$, $\alpha \approx 0.20$ (configurable).	TEU
String	$s_{ij} \in S$	Carrier service string this sailing belongs to	—
Period	$t_{ij} \in T$	Allocation period (month) this ETD falls in	—
Mean transit	μ_{ij}^{OC}	Mean transit days, POL to POD	days
Std dev	$\sigma_{ij}^{\mathrm{OC}}$	Std dev of transit	days
Carrier	$\mathrm{carrier}_{ij}$	Carrier name (e.g., MSC, CMA CGM, COSCO)	—
Vessel	vessel_{ij}	Vessel name	—

All-in per-FEU cost (derived):

$$c_{ij}^{\mathrm{FEU}} \triangleq r_{ij}^{\mathrm{FEU}} + \mathrm{BAF}_{ij} + \mathrm{THC}_{ij}^{\mathrm{POL}} + \mathrm{THC}_{ij}^{\mathrm{POD}}$$

Similarly c_{ij}^{TEU} uses r_{ij}^{TEU} in place of r_{ij}^{FEU}.

4.4 Dwell Arc Parameters

For each $(p_A, p_E) \in A_{\mathrm{DW}}$:

Parameter	Symbol	Description	MVP value
Mean dwell	μ_p^{DW}	Unloading + customs clearance (no exam)	USLAX/USLGB: 3.5 d; USSEA: 2.5 d
Std dev	σ_p^{DW}	Std dev of total dwell	USLAX/USLGB: 1.5 d; USSEA: 1.0 d
Free time	f_p	Free days before storage charges	5 d

P1: dwell becomes a function of hs_k, C-TPAT status, origin country, consignee history. See Section 13.

4.5 Inland Arc Parameters

For each $(p_E, h) \in A_{\mathrm{IL}}$:

Parameter	Symbol	Description	Unit
Cost	$c_{p_E h}^{\mathrm{IL}}$	Flat rate per container move (one chassis per container)	USD/move
Mean transit	$\mu_{p_E h}^{\mathrm{IL}}$	Mean days, POD exit to destination	days
Std dev	$\sigma_{p_E h}^{\mathrm{IL}}$	Std dev of transit	days
Mode	$\mathrm{mode}_{p_E h}$	FTL truck or intermodal rail	—
Road distance	$\mathrm{dist}_{p_E h}^{\mathrm{IL}}$	Informational	km

4.6 Carrier Allocation Parameters (String-Level)

Ocean carriers sell capacity in blocks on *service strings*. A string is a fixed port-call loop (e.g., MSC Tiger TPEB: SHA → NGB → SZX → USLAX → USLGB). A forwarder's contracted allocation covers *all* (POL, POD) pairs on a string — not a specific port pair.

For each string $s \in S$ and period $t \in T$:

Parameter	Symbol	Description	Unit
Contracted allocation	$\mathrm{alloc}(s, t)$	Forwarder's contracted block on string s in period t	TEU
Current utilization	$\mathrm{util}(s, t)$	TEU already booked/confirmed on s in t at solve time	TEU
Remaining capacity	$\mathrm{rem}(s, t)$	$\mathrm{alloc}(s, t) - \mathrm{util}(s, t)$; input to optimizer	TEU
Vessel cap fraction	α	Fraction of string allocation used as per-sailing slot cap proxy. Default 0.20.	—

$\mathrm{rem}(s, t)$ is computed before each solve from the shipment state store. The optimizer sees only remaining capacity — utilization is an external state input, not a decision variable.

Remark 3 (String vs. port-pair capacity)**.** A forwarder's contracted block on MSC Tiger (e.g., 400 TEU slots/month) applies to *all sailings* of that string in the month, across *all* POL/POD combinations. A booking on SHA→USLAX and a booking on NGB→USLGB both draw from the same pool.

Remark 4 (Unit convention)**.** All allocation quantities — $\mathrm{alloc}(s,t)$, $\mathrm{util}(s,t)$, $\mathrm{rem}(s,t)$ — are in **TEU slots**, consistent with $\mathrm{slots}(k,ij)$. Real carrier BSA contracts may be quoted in FEU (common on TPEB) or TEU depending on carrier and trade lane. Convert to TEU slots at input time: $\mathrm{alloc}_{\mathrm{FEU}} \times 2 = \mathrm{alloc}$ in TEU slots.

4.7 Pre-Computed Container Mix

For each commodity k and each feasible ocean arc $(i,j) \in A^k_{\mathrm{OC}}$, we pre-compute the optimal container mix that minimizes ocean cost while satisfying volume and weight constraints:

$$
\begin{aligned}
(f^*_k, t^*_k)\big|_{ij} = \arg \min_{f,\,t\,\geq 0,\ \text{integer}} \quad & f \cdot c^{\mathrm{FEU}}_{ij} + t \cdot c^{\mathrm{TEU}}_{ij} \\
\text{s.t.} \quad & 76f + 33t \geq v_k \quad \text{(volume)} && (2) \\
& 26{,}500f + 24{,}000t \geq w_k \quad \text{(weight)} && (3) \\
& f, t \in \mathbb{Z}_{\geq 0}
\end{aligned}
$$

This is a 2-variable integer program, solvable analytically (see Section 4.7). The results feed into the main MILP as parameters:

Derived parameter	Expression	Description
$\mathrm{cost}(k,ij)$	$f^*_k\big\vert_{ij} \cdot c^{\mathrm{FEU}}_{ij} + t^*_k\big\vert_{ij} \cdot c^{\mathrm{TEU}}_{ij}$	Optimal ocean cost for commodity k on sailing (i,j)
$\mathrm{slots}(k,ij)$	$2f^*_k\big\vert_{ij} + t^*_k\big\vert_{ij}$	TEU slots consumed on the vessel

Remark 5 (n_k and physical container count)**.** n_k counts FEU-sized moves and is used to price the land legs: n_k trucks at rate c^{PC} or c^{IL}, one chassis per container. On ocean arcs, the mix pre-computation may assign a different combination of FEUs and TEUs. The total physical container count on sailing (i,j) is $f^*_k\big|_{ij} + t^*_k\big|_{ij}$, which is always $\geq n_k$ (you cannot cover the same cargo with fewer physical containers than the FEU minimum). The gap is at most 1:

$$
\left(f^*_k\big|_{ij} + t^*_k\big|_{ij}\right) - n_k \leq 1
$$

This holds because replacing one FEU with two or more TEUs is never cost-effective at $\rho \in (0.56, 0.86)$: three TEUs cost $3\rho \geq 1.68\, c^{\mathrm{FEU}}_{ij}$ but cover only 99 CBM, less volume per dollar than one FEU. The only viable substitution is replacing the single residual FEU with one TEU when the remaining volume fits (≤ 33 CBM), which leaves the total count unchanged at n_k. In practice the deviation is zero; the bound ≤ 1 is a theoretical guarantee.

Container mix analytical solution. Given cost ratio $\rho = c^{\mathrm{TEU}}_{ij}/c^{\mathrm{FEU}}_{ij} \in (0.56, 0.86)$ (see rate ratio table above) and constraints (2)–(3):

1. Compute the minimum FEU count needed from volume and weight: $f_v = \lceil v_k/76 \rceil$, $f_w = \lceil w_k/26{,}500 \rceil$. Set $f_{\min} = \max(f_v, f_w)$.
2. For each candidate $f \in \{f_{\min}, f_{\min} - 1, \ldots, 0\}$:

a. $t_v = \max(0, \lceil (v_k - 76f)/33 \rceil)$ (minimum TEUs to cover remaining volume).
b. $t_w = \max(0, \lceil (w_k - 26{,}500f)/24{,}000 \rceil)$ (minimum TEUs to cover remaining weight).
c. $t = \max(t_v, t_w)$.
d. Evaluate $C(f, t) = f \cdot c_{ij}^{\text{FEU}} + t \cdot c_{ij}^{\text{TEU}}$.

3. Return $(f^*, t^*) = \arg\min C(f, t)$ over all candidates from step 2.

The enumeration runs over at most $f_{\min} + 1$ candidates, which is small in practice (typical $f_{\min} \le 5$). Since $\rho \in (0.63, 0.83)$, one FEU covers $76/33 \approx 2.3$ TEU-equivalents of volume at a cost of $1/\rho \in (1.20, 1.59)$ TEU-equivalents — so FEU is always cheaper per CBM. The optimal solution is therefore $(f_{\min}, 0)$ or $(f_{\min} - 1, t)$ for small residual t, but explicit enumeration guarantees optimality without relying on this structural property.

5 Commodity-Specific Subgraph Construction

Before the MILP is built, we construct $G(N_k, A_k)$ for each commodity k. The subgraph contains only arcs that are (i) time-feasible and (ii) reachable on a complete path from $o(k)$ to $d(k)$. Any arc not on such a path is excluded.

Key structural observation. In this network, each commodity k has *at most one pre-carriage arc per POL*: the arc $(o(k), j)$ for each $j \in N_{\text{POL}}$. The transit time $\mu_{o(k),j}^{\text{PC}}$ is therefore unambiguous for a given (commodity, POL) pair. This eliminates the arc-pairing ambiguity that would arise if multiple different pre-carriage routes from different origins could reach the same POL with different transit times.

Construction algorithm.

1. **Pre-carriage pass.** For each POL $i \in N_{\text{POL}}$, compute earliest arrival:

$$\tau_k(i) \triangleq t_k^{\text{rdy}} + \mu_{o(k),i}^{\text{PC}}$$

   Arc $(o(k), i)$ is *candidate* iff $\exists\,(i, j) \in A_{\text{OC}}$ such that $\tau_k(i) \le \text{CYC}_{ij}$ (can make at least one sailing from i).

2. **Ocean sailing pass.** For each arc $(i, j) \in A_{\text{OC}}$:

$$(i, j) \in A_{\text{OC}}^k \iff \tau_k(i) \le \text{CYC}_{ij} \text{ and } \text{ETD}_{ij} + \mu_{ij}^{\text{OC}} + \mu_p^{\text{DW}} + \mu_{p_E,d(k)}^{\text{IL}} \le T_k^{\text{dead}}$$

   where $j \in N_{\text{POD},A}$, p is the physical port at j, and p_E is its exit node. The first condition is the CY cutoff check (directly against commodity k's pre-carriage transit to POL i — no ambiguity). The second is the end-to-end deadline check.

3. **Dwell pass.** Include dwell arc (p_A, p_E) iff $\exists\,(i, j) \in A_{\text{OC}}^k$ with $j = p_A$ (i.e., some feasible sailing arrives at that POD).

4. **Inland pass.** For each POD exit p_E and inland arc $(p_E, h) \in A_{\text{IL}}$:

$$(p_E, h) \in A_{\text{IL}}^k \iff h = d(k) \text{ and } \exists\,(i, j) \in A_{\text{OC}}^k : \text{ETD}_{ij} + \mu_{ij}^{\text{OC}} + \mu_p^{\text{DW}} + \mu_{p_E h}^{\text{IL}} \le T_k^{\text{dead}}$$

5. **Trim pre-carriage.** Remove any pre-carriage arc $(o(k), i)$ for which no ocean arc $(i, j) \in A_{\text{OC}}^k$ remains adjacent to i.

6. **Reachability sweep.** Run a forward BFS from $o(k)$ and a backward BFS from $d(k)$ on the candidate arc set. Retain only arcs on paths reachable from both ends. Result: $G(N_k, A_k)$ where every arc lies on at least one complete $o(k) \to d(k)$ path.

Pre-filter scope guard. *[Note H3: The pre-filter only processes (origin, destination) pairs whose trade lane is covered by the configured carrier services. O-D pairs outside the configured scope must be hard-rejected before the pre-filter runs, with a structured error returned to the agent.]*

[Note H4: Hazmat, OOG, and reefer commodities ($\tau_k \neq$ general) must be rejected at the pre-filter boundary with a structured infeasibility report. Routing them identically to general cargo produces silently incorrect results. The instance generator already restricts to general cargo (see §4.1 Remark); enforce the same check at runtime.]

[Note H2: The deadline check in step 2 uses mean transit times. When a sailing's margin is tight, mean-based inclusion overstates on-time probability. P1: add a safety buffer $\delta \approx 1$–2 d to the deadline check: $ETD_{ij} + \mu^{OC}_{ij} + \mu^{DW}_{p} + \mu^{IL}_{p_E,d(k)} + \delta \leq T^{dead}_k$. The CYC risk note in Open Item 3 partially addresses this for the pre-carriage side.]

Infeasible commodities. If $A^k = \emptyset$ after the above (no feasible path exists), commodity k is *removed from K before the MILP is built.* The pre-solve returns a structured infeasibility report:

`{commodity_id, reason: "no_feasible_path", earliest_achievable_days, deadline_shortfall_days}`

The agent surfaces this to the user with a recommended recovery action (relax deadline, consider air mode). The MILP is never passed an infeasible commodity.

6 Decision Variables

$$x^k_{ij} \in \{0,1\} \qquad \forall\, k \in K,\ (i,j) \in A^k$$

$x^k_{ij} = 1$ if commodity k is routed on arc (i,j).

Remark 6 (No-split semantics)**.** $x^k_{ij} = 1$ means *all* containers for commodity k traverse arc (i,j) together. Commodity splitting across multiple sailings or paths is prohibited. For an ocean arc, this means all $f^*_k\big|_{ij}$ FEUs and $t^*_k\big|_{ij}$ TEUs depart on the same vessel. This reflects standard FCL booking practice: one booking covers all containers for a shipment.

7 Objective Function

$$\min \quad Z = \sum_{k \in K} \left[\underbrace{\sum_{(i,j) \in A^k_{\mathrm{PC}}} n_k \cdot c^{\mathrm{PC}}_{ij} \cdot x^k_{ij}}_{\text{pre-carriage}} + \underbrace{\sum_{(i,j) \in A^k_{\mathrm{OC}}} \mathrm{cost}(k, ij) \cdot x^k_{ij}}_{\text{ocean (optimal mix)}} + \underbrace{\sum_{(p_E,h) \in A^k_{\mathrm{IL}}} n_k \cdot c^{\mathrm{IL}}_{p_E h} \cdot x^k_{p_E h}}_{\text{inland}} \right] \tag{4}$$

Cost structure.

- *Pre-carriage:* n_k trucks (one chassis per container), each at flat rate c^{PC}_{ij}.
- *Ocean:* $\mathrm{cost}(k, ij)$ is the pre-computed optimal cost for commodity k on sailing (i,j), accounting for the optimal FEU/TEU mix. Dwell cost (port storage beyond free time) deferred to P1.

- *Inland:* n_k chassis moves, each at flat rate $c^{\mathrm{IL}}_{p_E h}$ (one chassis per container for FCL drayage).

[Note C1: Pre-carriage and inland costs use n_k as the container-count multiplier. The actual count on the chosen sailing is $f^*_k\big|_{ij} + t^*_k\big|_{ij}$. Because $2\rho > 1$ for all $\rho \in (0.56, 0.86)$, the optimizer never replaces a FEU with 2 TEUs, so $f^* + t^* = n_k$ in every feasible solution. Using n_k is therefore exact, not an approximation.]

8 Constraints

8.1 Flow Conservation

For each commodity $k \in K$ and node $n \in N_k$:

$$\sum_{(n,j)\in A^k} x^k_{nj} \;-\; \sum_{(i,n)\in A^k} x^k_{in} \;=\; b_k(n) \qquad \forall\, k \in K,\; n \in N_k \tag{5}$$

$$b_k(n) = \begin{cases} +1 & n = o(k) \quad \text{(net outflow = 1 at source)} \\ -1 & n = d(k) \quad \text{(net inflow = 1 at sink)} \\ 0 & \text{otherwise} \end{cases}$$

Flow conservation ensures each commodity routes along exactly one complete path (one arc of each type: PC, OC, DW, IL). The subgraph construction guarantees any arc in A^k is on a valid complete path, so feasibility is structurally enforced — no additional time constraints are needed in the MILP.

8.2 Vessel Capacity Constraint

For each ocean arc $(i, j) \in A_{\mathrm{OC}}$:

$$\sum_{k\in K:\,(i,j)\in A^k_{\mathrm{OC}}} \mathrm{slots}(k, ij)\cdot x^k_{ij} \;\le\; \mathrm{cap}^{\mathrm{TEU}}_{ij} \qquad \forall\,(i,j) \in A_{\mathrm{OC}} \tag{6}$$

The total TEU slots consumed on a single sailing cannot exceed that sailing's slot availability. Without this constraint, the string allocation constraint alone allows all monthly forwarder allocation to concentrate on one departure, which is physically impossible.

Proxy for $\mathbf{cap}^{\mathbf{TEU}}_{ij}$. Real per-sailing slot availability is not directly observable by a forwarder. As a non-binding placeholder:

$$\mathrm{cap}^{\mathrm{TEU}}_{ij} \triangleq \alpha \cdot \mathrm{alloc}(s_{ij},\, t_{ij})$$

$\alpha \approx 0.20$ (configurable). Rationale: with ≈ 4 sailings per month per string, $\alpha = 0.20$ caps any single departure at roughly 80 % of the full monthly contracted block — large enough to be non-binding on typical MVP instances, tight enough to prevent degenerate concentration. See Open Item 7 for the path to a real estimate.

8.3 String Allocation Constraint

For each carrier service string $s \in S$ and time period $t \in T$:

$$\sum_{\substack{(i,j)\in A_{\mathrm{OC}}\\ s_{ij}=s,\ t_{ij}=t}} \ \sum_{k\in K:\,(i,j)\in A^k_{\mathrm{OC}}} \mathrm{slots}(k,ij)\cdot x^k_{ij} \ \le\ \mathrm{rem}(s,t) \qquad \forall\, s\in S,\ t\in T \tag{7}$$

The total TEU slots consumed across *all sailings of string s in period t* cannot exceed the forwarder's remaining allocated capacity. This constraint couples routing decisions across all (POL, POD) pairs on the same string within the same month.

8.4 Budget Cap (Optional)

For each $k \in K$ with finite B_k:

$$\sum_{(i,j)\in A^k_{\mathrm{PC}}} n_k c^{\mathrm{PC}}_{ij} x^k_{ij} + \sum_{(i,j)\in A^k_{\mathrm{OC}}} \mathrm{cost}(k,ij)\, x^k_{ij} + \sum_{(p_E,h)\in A^k_{\mathrm{IL}}} n_k c^{\mathrm{IL}}_{p_E h} x^k_{p_E h} \le B_k \tag{8}$$

8.5 Variable Domains

$$x^k_{ij} \in \{0,1\} \qquad \forall\, k\in K,\ (i,j)\in A^k \tag{9}$$

9 Complete Formulation

$$\min \quad Z = \sum_{k\in K}\left[\sum_{(i,j)\in A^k_{\mathrm{PC}}} n_k c^{\mathrm{PC}}_{ij} x^k_{ij} + \sum_{(i,j)\in A^k_{\mathrm{OC}}} \mathrm{cost}(k,ij)\, x^k_{ij} + \sum_{(p_E,h)\in A^k_{\mathrm{IL}}} n_k c^{\mathrm{IL}}_{p_E h} x^k_{p_E h}\right] \tag{P}$$

$$\text{s.t.} \quad \sum_{(n,j)\in A^k} x^k_{nj} - \sum_{(i,n)\in A^k} x^k_{in} = b_k(n) \qquad \forall\, k\in K,\ n\in N_k \tag{P.1}$$

$$\sum_{k\in K:\,(i,j)\in A^k_{\mathrm{OC}}} \mathrm{slots}(k,ij)\, x^k_{ij} \le \mathrm{cap}^{\mathrm{TEU}}_{ij} \qquad \forall\, (i,j)\in A_{\mathrm{OC}} \tag{P.2}$$

$$\sum_{\substack{(i,j)\in A_{\mathrm{OC}}\\ s_{ij}=s,\, t_{ij}=t}} \ \sum_{k:\,(i,j)\in A^k_{\mathrm{OC}}} \mathrm{slots}(k,ij)\, x^k_{ij} \le \mathrm{rem}(s,t) \qquad \forall\, s\in S,\ t\in T \tag{P.3}$$

(P.4): budget constraints (8) for commodities with finite B_k

$$x^k_{ij} \in \{0,1\} \qquad \forall\, k\in K,\ (i,j)\in A^k \tag{P.5}$$

Remark 7 (Problem structure)**.** (P) is a Binary Multi-Commodity Network Flow (BMCNF). The LP relaxation of a pure MCF is TU; coupling via the string allocation constraint (P.3) breaks TU, making integrality non-guaranteed. For small MVP instances ($|K| \le 50$), HiGHS solves to optimality easily. At forwarder scale ($|K| \ge 500$), LP bound quality and symmetry breaking become relevant (see Section 12.4).

10 Graph Decomposition for Independent Subproblems

When routing a mixed batch of commodities (e.g., from both TPEB and FEWB trade lanes), groups of commodities may share no supply arcs and no allocation pools — making their routing decisions completely independent. Identifying and solving these independently reduces MILP solve time and enables parallelism.

10.1 Commodity-Supply Graph

Define a commodity-supply graph $H = (K, E_H)$:

- **Nodes:** one per commodity $k \in K$.
- **Edges:** $(k_1, k_2) \in E_H$ iff commodities k_1 and k_2 share supply:
  1. $\exists\,(i,j) \in A_{\mathrm{OC}}^{k_1} \cap A_{\mathrm{OC}}^{k_2}$ (same feasible sailing), *or*
  2. $\exists\, s \in S,\ t \in T$ such that both k_1 and k_2 have a feasible sailing on string s in period t: i.e., $\exists\,(i,j) \in A_{\mathrm{OC}}^{k_1}$ with $s_{ij} = s,\ t_{ij} = t$, and $\exists\,(i',j') \in A_{\mathrm{OC}}^{k_2}$ with $s_{i'j'} = s,\ t_{i'j'} = t$ (same allocation pool). Arc membership in A_{OC}^{k} is determined by the subgraph pre-filter (Section 5), not the MILP solve itself.

10.2 Decomposition Algorithm

1. Build H from $\{A_{\mathrm{OC}}^{k}\}_{k\in K}$ and the sailing-to-string mapping s_{ij}.
2. Find connected components $\mathcal{C}_1, \mathcal{C}_2, \ldots$ of H.
3. For each component $\mathcal{C}_i$: solve the MILP restricted to $K_i = \mathcal{C}_i$.
4. Merge solutions.

[Note M9: rem(s,t) must be computed once before the decomposition and held fixed for all component solves. Updating it between components — e.g., subtracting slots used by a solved component before solving the next — is incorrect: the allocation constraint in each component references the full remaining capacity at solve time, and components are, by construction, independent.]

Example. TPEB commodities (SHA→USLAX on MSC Tiger) and FEWB commodities (SHA→NLRTM on MSC Silk) use different strings with separate allocation pools and no overlapping sailings. They form disconnected components in H and are solved independently. Within TPEB, two commodities that both have SHA→USLAX in their feasible arc sets are connected in H (shared sailing, shared allocation) and solved jointly.

11 Transit Time Estimation

Transit times are used in the subgraph pre-filter and by the instance generator. They are parameters in MVP; in P1 they become distribution parameters for chance constraints.

11.1 Trucking (Pre-Carriage and Inland)

Geodesic distance (Haversine):

$$d_{ij}^{\mathrm{geo}} = 2R \arcsin\sqrt{\sin^2 \tfrac{\phi_j - \phi_i}{2} + \cos\phi_i \cos\phi_j \sin^2 \tfrac{\lambda_j - \lambda_i}{2}}, \quad R = 6{,}371\,\mathrm{km}$$

Road distance:

$$\text{dist}_{ij}^{\text{road}} = \kappa \cdot d_{ij}^{\text{geo}}, \quad \kappa = \begin{cases} 1.25 & \text{China pre-carriage} \\ 1.20 & \text{US inland} \end{cases}$$

Transit time:

$$\mu_{ij}^{\text{truck}} = \frac{\text{dist}_{ij}^{\text{road}}}{v^{\text{truck}}}, \quad v^{\text{truck}} = \begin{cases} 600\,\text{km/day} & \text{China (urban congestion, regulations)} \\ 800\,\text{km/day} & \text{US OTR (HOS 11-hour limit)} \end{cases}$$

Std dev: $\sigma_{ij}^{\text{truck}} = 0.15\,\mu_{ij}^{\text{truck}}$ (15% CV).

Transit time range (for robust model, P1): Each arc carries $[\mu - 2\sigma,\ \mu + 2\sigma]$ as a 95% interval. The instance generator outputs both mean and std dev; the range is the hook for P1 robustness.

11.2 Ocean Sailing

Sailing distance:

$$\text{dist}_{ij}^{\text{sail}} = 1.15 \cdot d_{ij}^{\text{geo}} \quad \text{(Trans-Pacific lane factor)}$$

[Note H1: 1.15 is calibrated for TPEB. For FEWB and TAWB, the factor is approximately 1.06 (shorter great-circle deviation on the Europe–Asia and transatlantic lanes). P1: replace the scalar with a per-arc or per-trade-lane parameter γ_{ij}.]

Transit time:

$$\mu_{ij}^{\text{OC}} = \frac{\text{dist}_{ij}^{\text{sail}}}{600\,\text{km/day}} \quad (\approx 13.5\,\text{knots average, inclusive of port approach and anchorage})$$

Validation: SHA→USLAX geodesic ≈ 9,200 km. Sailing distance ≈ 10,580 km. Transit ≈ 17.6 days. Published carrier schedule for this lane: 14–18 days. Estimate is within range.

Std dev: $\sigma_{ij}^{\text{OC}} = 0.10\,\mu_{ij}^{\text{OC}}$ (10% CV).

12 Instance Generator Specification

The instance generator produces synthetic but geographically and commercially realistic problem instances for solver testing. The generator is demand-first: commodity requests are generated first, then supply (sailings) is built to cover that demand at a configurable capacity utilization level.

Note on build process. The generator implementation is a joint session between the developer and the system designer. Do not implement independently.

12.1 Configuration Schema

All instance parameters are controlled by a `GeneratorConfig` object:

```
GeneratorConfig:

  # Trade lanes (drives POL/POD sets, carrier services, realistic schedules)
  trade_lanes: List[str]         # e.g. ["TPEB", "FEWB", "TAWB"]

  # Demand
  n_commodities: int             # total commodities to generate
  start_date: date               # scenario anchor date
  date_horizon_days: int         # window over which cargo-ready dates spread
  cargo_ready_spread_days: int   # subset of horizon for cargo-ready sampling

  # Commodity profile
  volume_range_cbm: [float, float]      # [min, max]
  weight_range_kg: [float, float]       # [min, max] | upper end tests weight-limited n_k
  deadline_slack_days: [int, int]       # [min, max] days added beyond fastest feasible path
  cargo_type_mix: dict                  # {general: 0.80, hazmat: 0.10, reefer: 0.10}
                                        # MVP: all routed as general; mix recorded only

  # Supply calibration
  load_factor: float             # target (total TEU demand) / (total TEU capacity)
                                 # 0.5 = easy, 0.8 = realistic tight, >1.0 = capacity-infeas
  target_infeasible_fraction: float  # fraction of commodities with no feasible path
                                     # injected deliberately; tests infeasibility handling
  sailings_per_week_per_string: int  # 1 = weekly service (standard)
  n_carriers_per_lane: int           # competing carriers per trade lane

  # Carrier allocation (string-level)
  allocation_per_service_month_teu: int  # forwarder contracted block per string per month
  initial_utilization_fraction: float   # fraction already booked at solve time
                                        # rem = alloc * (1 - initial_utilization_fraction)
  target_alloc_utilization_fraction: float  # target fraction of alloc consumed at solve opt
                                            # used to calibrate alloc vs cap (see step 5)
  vessel_cap_alpha: float               # alpha in cap_{ij} = alpha * alloc(s,t); default 0.

  # Container cost (market-calibrated defaults)
  feu_base_rate_range_usd: [float, float]   # [min, max], e.g. [2000, 6000]
  teu_feu_cost_ratio: float                 # TEU rate = FEU rate * ratio; TPEB: ~0.79, FEWB
  feu_hc_premium_usd: float                 # surcharge for 40'HC vs 40' standard; ~$50-150
  baf_fraction: float                       # BAF as fraction of base rate; ~0.15-0.25
  thc_pol_range_usd: [float, float]         # per container
  thc_pod_range_usd: [float, float]         # per container

  # Transit time
  transit_sigma_fraction: float   # sigma = fraction * mean for all arc types; default 0.12
                                  # [Note M5: Section 9 specifies 0.15 for trucking and 0.10
                                  #  for ocean. This config field is a single override. P1:
                                  #  split into trucking_sigma_fraction and ocean_sigma_frac
  vessel_speed_kmday: float       # default 600 (~13.5 knots)

  # Random seed
  random_seed: int
```

12.2 Trade Lane Definitions

Code	Direction	POL ports	POD ports
TPEB	Asia → US West	SHA, NGB, SZX	USLAX, USLGB, USSEA
FEWB	Asia → Europe	SHA, NGB, SZX, SGSIN	NLRTM, DEHAM, GBFXT, BEFBR
TAWB	Europe → US East	NLRTM, DEHAM, GBFXT, BEFBR	USNYK, USSAV, USBAL, USORF
TPWB	US West → Asia	USLAX, USLGB, USSEA	SHA, NGB, SZX

12.3 Demand-First Generation Algorithm

1. **Generate commodities.**
   (a) Sample origin city from cities near the POLs of the configured trade lanes.
   (b) Sample destination city from cities near the PODs.
   (c) Sample $v_k \sim \text{Uniform}(v_{\min}, v_{\max})$, $w_k \sim \text{Uniform}(w_{\min}, w_{\max})$.
   (d) Compute $n_k = \max(\lceil v_k/76 \rceil, \lceil w_k/26{,}500 \rceil)$.
   (e) Sample t_k^{rdy} uniformly over `[start_date, start_date + cargo_ready_spread_days]`.

2. **Compute minimum feasible time for each commodity.**
   (a) For each (origin, POL) pair: compute pre-carriage transit $\mu^{\text{PC}}_{o(k),j}$.
   (b) Add CY buffer (4 days).
   (c) Add fastest ocean transit to any reachable POD.
   (d) Add port dwell.
   (e) Add fastest inland transit to destination.
   (f) min_time_k = minimum over all (origin → POL → POD → destination) paths.

3. **Set deadlines.**
   - Normal commodities: $T_k^{\text{dead}} = t_k^{\text{rdy}} + \text{min\_time}_k + \text{Uniform}(\text{deadline\_slack\_min}, \text{deadline\_slack\_max})$.
   - Infeasible commodities (fraction = `target_infeasible_fraction`): $T_k^{\text{dead}} = t_k^{\text{rdy}} + 0.7 \times \text{min\_time}_k$ (impossible).

4. **Generate sailings (demand-driven).**
   (a) For each (POL, POD, trade lane) pair, determine the cargo-ready date window requiring coverage: $[t_k^{\text{rdy}}, t_k^{\text{rdy}} + \max \mu^{\text{PC}}]$ across all commodities.
   (b) Generate weekly sailings (one per string per week) covering this window.
   (c) Assign realistic carrier, service, and vessel names from the trade lane table.
   (d) Compute μ_{ij}^{OC} and σ_{ij}^{OC} via Section 11.
   (e) Set $\text{CYC}_{ij} = \text{ETD}_{ij} - 4\,\text{days}$. *Sync note: this 4-day buffer must match the value used in step 2b of the minimum-feasible-time computation. Both occurrences must be updated together if per-port CYC variation is introduced (P1 extension).*
   (f) **Calibrate per-sailing capacity to hit `load_factor`:** $\text{cap}_{ij}^{\text{TEU}} = \text{round}(\text{total TEU demand on this}$ `load_factor`). *[Note C3: Circular dependency — cap_{ij}^{TEU} depends on total TEU demand, which depends on which sailings are feasible (step 4a/b), which depends on schedule coverage. Resolve by computing demand in step 1 first (with n_k container counts), then setting cap. Do not re-run step 1 after setting cap.]*

5. **Set allocation state.** $\text{rem}(s,t) =$ `allocation_per_service_month_teu`$\times(1-$`initial_utilization_fra`
   Calibration: set `allocation_per_service_month_teu` $\approx \left(\sum_{(i,j):\, s_{ij}=s,\, t_{ij}=t} cap_{ij}^{TEU}\right) \times$ `target_alloc_util`
   so that the string allocation constraint (P.3) binds at the intended load level.

6. **Pre-compute container mix.** For each (commodity, feasible sailing) pair, run the mix algorithm (Section 4.7) to get f_k^*, t_k^*, $\mathrm{cost}(k, ij)$, $\mathrm{slots}(k, ij)$.
7. **Run subgraph construction** (Section 5). Log infeasible commodities. Verify infeasible fraction is within 20% of target; adjust `deadline_slack` if needed and re-run.
8. **Run decomposition** (Section 10). Log connected components and their sizes. This verifies that TPEB/FEWB mixed instances decompose as expected.
9. **Serialize** to JSON with schema matching parameter tables above. Include random seed and config for reproducibility.

12.4 Solver Concerns

- **Instance size:** At $|K| = 50$, ≈ 72 arcs, and ≈ 30 feasible arcs per commodity, total binary variables $\approx 1{,}500$. Trivial for HiGHS.
- **Symmetry:** Inject small random perturbations ($\pm 1\%$) to FEU rates to break symmetry between near-identical sailings on the same lane.
- **LP bound:** The string allocation constraint (P.3) is the primary source of LP bound weakness. For tight-capacity instances, the gap may require B&B. HiGHS handles this well for MVP sizes. *[Note M4: The gap bound $\left(f_k^*\big|_{ij} + t_k^*\big|_{ij}\right) - n_k \le 1$ in §4.5 Remark is proven by a cost argument: $2\rho > 1$ for all $\rho > 0.5$, so the optimizer never replaces 1 FEU with 2 TEUs. The current remark text mentions "three TEUs" which is the wrong comparison unit — the binding case is 2 TEUs vs 1 FEU. Rephrase in a future edit.]*

13 Deferred: P1 Extensions

13.1 Commodity-Specific Customs Inspection Model (P1)

Dwell arc weight becomes a function of commodity attributes:

$$\mu_k^{\mathrm{DW}}(p) = \mu_p^{\mathrm{unload}} + t^{\text{no-exam}} + p_k^{\mathrm{exam}}(p) \cdot t^{\mathrm{exam}}$$

$$p_k^{\mathrm{exam}}(p) = \mathrm{clip}\left(\lambda_p^{\mathrm{base}} \cdot \eta_k^{\mathrm{HS}} \cdot \eta_k^{\mathrm{orig}} \cdot (1 - \delta_k^{\mathrm{CTPAT}}),\ 0,\ 1\right)$$

13.2 Time-Phased Capacity Release (P1)

Even with $\mathrm{rem}(s, t)$ available, a forwarder may not want to commit all remaining capacity in one routing run — future urgent shipments may need it. This requires forecasting future demand and releasing capacity in tranches:

$$\mathrm{open}(s, t, \ell) \le \mathrm{rem}(s, t)$$

where ℓ is the routing run (batch) index and $\mathrm{open}(s, t, \ell)$ is the capacity released for run ℓ. Requires a demand forecast model. Deferred.

13.3 Probabilistic Delivery Constraint (P1)

Replace deterministic deadline feasibility with:

$$\Pr\left(\text{delivery time}_k \le T_k^{\mathrm{dead}}\right) \ge \alpha_k$$

where $\alpha_k \in \{0.80, 0.90, 0.95\}$ maps to service level (Economy/Standard/Express).

13.4 Other P1 Items

TEU-only container option for heavy cargo; multi-objective Pareto frontier (cost vs. transit time vs. on-time probability); port storage cost in objective; carrier allocation caps as soft constraints with penalty; transshipment arcs; multi-HS-code commodity schema (real FCL bookings can consolidate multiple products with different HS codes — MVP uses a single hs_k field, which is harmless while the customs model is deferred; P1 should replace hs_k with a list and derive inspection risk as the maximum risk tier across all codes in the shipment).

14 Open Items

1. **String allocation granularity:** Current model uses monthly periods for allocation. Some carrier contracts use 13 four-week periods per year rather than calendar months. Confirm with design partners before implementing.

2. **Multi-POL / transshipment strings:** Some strings have calls at multiple Chinese ports (e.g., SHA $\rightarrow$ NGB $\rightarrow$ SZX before crossing). A booking on SHA$\rightarrow$USLAX and NGB$\rightarrow$USLAX on the same sailing are not independent — they're on the same vessel. The current model treats them as independent sailings, which may overstate available capacity. Deferred to when transshipment arcs are added in P1.

3. **CYC compliance risk:** The subgraph construction uses mean transit times. A pre-carriage arc with mean transit exactly at the CYC has $\approx 50\%$ rollover risk. Flag this in the agent's output when the margin is < 0.5 days.

4. **Infeasibility report schema:** Define the JSON schema for the infeasibility report before coding the pre-filter. Minimum fields: `commodity_id, reason, earliest_achievable_days, deadline_shortfall_days, nearest_feasible_deadline`.

5. **Generator: sailing schedule realism:** Real carrier schedules have port-specific cutoff windows, vessel-rotation identifiers, and multi-port calls. The generator should reproduce these from public schedule data, not fabricate them.

6. **BSA contract unit convention:** The model uses TEU slots internally for all allocation quantities ($\mathrm{alloc}(s,t)$, $\mathrm{util}(s,t)$, $\mathrm{rem}(s,t)$). Real BSA contracts on TPEB are commonly quoted in FEU; other trade lanes may use TEU or gross tonnage. Confirm with design partner: (a) are contracts quoted in FEU or TEU on the primary TPEB lanes, and (b) is the $\times 2$ conversion ($\mathrm{alloc}_{\mathrm{FEU}} \times 2 = \mathrm{alloc}$ in TEU slots) standard across all contract types? See also the unit convention remark in Section 4.6.

7. **Vessel capacity proxy (P.2):** The per-sailing cap $\mathrm{cap}_{ij}^{\mathrm{TEU}} = \alpha \cdot \mathrm{alloc}(s_{ij}, t_{ij})$ with $\alpha = 0.20$ is a placeholder. The right estimate is the fraction of the vessel's actual TEU capacity reserved for this forwarder on this string — which requires carrier contract data or an AIS-based slot utilization feed. Candidate approaches: (a) request utilization reports directly from carriers in the BSA contract, (b) infer from AIS vessel identifiers and known vessel TEU capacities (publicly available), (c) use $\alpha = 0.5$ as a less conservative default once nominal carrier vessel sizes are known. *Until real data is available,* $\alpha = 0.20$ *keeps the constraint non-binding while providing the structural guard against degenerate solutions.*